

DURGESH A P

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EXPERIENCE

Mayonix Software Solutions | 🌐 mayonix.com

Aug 2025 – Aug 2026 (Expected)

Full-Stack Developer Intern

- Architected and developed a full-stack corporate website using Next.js 15, TypeScript, Strapi CMS v5, and Neon PostgreSQL in a decoupled monorepo architecture, delivering 8+ dynamic pages with SSR and ISR.
- Built a custom high-performance API middleware with Upstash Redis caching and lifecycle-hook-based auto-invalidation, reducing API response times by approximately 95% on cached content endpoints.
- Developed 25+ reusable React components with Tailwind CSS 4, including a CMS-driven dynamic navbar, rich-text blog renderer, portfolio showcase, FAQ system, and careers module with responsive layouts.
- Implemented end-to-end SEO infrastructure, including dynamic sitemap generation, robots.txt, Open Graph/Twitter Card meta tags, and JSON-LD structured data, ensuring search engine discoverability.
- Designed 8+ Strapi content types with Cloudinary media integration, transactional email workflows via Resend, and managed multiple CMS version upgrades while maintaining backward compatibility.

PROJECTS

Multimodal Ore Identification using Deep Learning

Jan 2026 – Present

Python, TensorFlow, Fuzzy Logic, Next.js, Raspberry Pi

- Designed a hybrid multimodal classification system fusing RGB, microscopic, infrared, acoustic, and capacitive sensor data for non-destructive identification of 5 mineral classes (Pyrite, Chalcopyrite, Hematite, Magnetite, Bornite).
- Proposed an ensemble CNN + Vision Transformer architecture with attention-based multimodal fusion, dual prediction heads for ore classification and elemental composition regression, and a fuzzy logic layer for uncertainty management.
- Architected an IoT edge-to-cloud deployment pipeline using Raspberry Pi for edge acquisition, a Dockerized TensorFlow backend with MLflow/Optuna, and a Next.js dashboard for real-time visualization and analytics.

Optimizing Drone-Truck Delivery using ALNS | 🏠 GitHub

Jun 2025 – Nov 2025

Python, FastAPI, PyQt5, ALNS

- Built a full-stack simulation platform (FastAPI, PyQt5) for last-mile delivery using a heterogeneous fleet of fuel trucks, electric trucks, and drones.
- Implemented an Adaptive Large Neighborhood Search (ALNS) algorithm to simultaneously minimize travel time, energy consumption, CO₂ emissions, and noise pollution.
- Validated the framework on datasets up to 500 customers, achieving a consistent mean optimization improvement of 21.1% and reducing operational costs.

Automated Job Discovery in CareerTech | 🏠 GitHub

Jan 2025 – Apr 2025

Next.js, Python, Scrapy, Selenium, PostgreSQL

- Built an automated job aggregation platform to streamline job data extraction, parsing, and intelligent filtering.
- Developed the full-stack interface and API backend using Next.js, integrating PostgreSQL for structured data storage.
- Engineered robust scraping pipelines using Python, Scrapy, and Selenium to extract real-time postings from dynamic career sites.

TECHNICAL SKILLS

Languages: Java, Python, TypeScript, R, C

Frameworks & Libraries: React, Next.js, Node.js, FastAPI, Strapi CMS, Tailwind CSS, Bootstrap

Databases: MongoDB, PostgreSQL, MySQL, SQLite, Redis

Developer Tools: Git, GitHub, Gradle, Jenkins, Postman, Prisma, Selenium, Scrapy, Power BI, LaTeX

Cloud & Platforms: Google Cloud Platform, Vercel, Cloudinary, Upstash, Neon, Linux

Design & Visualization: Figma, Canva, Matplotlib, Seaborn, Plotly

EDUCATION

A J Institute of Engineering & Technology

Aug 2023 – May 2027 (Expected)

B.E. in Computer Science & Engineering | CGPA: 8.63

Mangaluru, Karnataka

LEADERSHIP & EXTRACURRICULAR

- **Hackathon Success:** Secured **First Runner-Up** at Matrix Fusion 4.0 (24-Hour National Hackathon, 2026). Active participant in TetherX (VIT Chennai), Smart India Hackathon, and Hacktoberfest (2025).
- **Research & Publications:** Co-authored a research paper on ALNS-based Drone-Truck Delivery; currently under revision at the *International Journal of Modeling, Simulation, and Scientific Computing* (IJMSSC).
- **Leadership & Mentorship:** Student Coordinator for SIH 2024; organized UI/UX and Python workshops; conducted a 3-day hands-on DBMS workshop at Reva University, Bengaluru.